

PREDICTING 30-DAY HOSPITAL READMISSION FROM NOISY DIABETES RECORDS UNDER ADVERSARIAL CONDITIONS

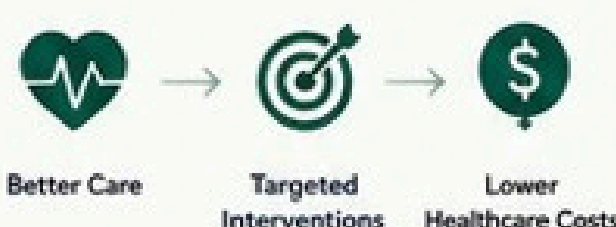
A Machine Learning Approach to Improve Patient Outcomes and Reduce Healthcare Costs

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1. INTRODUCTION

Hospital readmission within 30 days is a critical quality indicator and a significant financial burden on healthcare systems. Accurate prediction helps identify high-risk patients, enabling targeted interventions that can improve outcomes and reduce unnecessary readmissions.



2. PROBLEM STATEMENT

Severe class imbalance:
≈ 91% non-readmitted vs.
≈ 9% readmitted

Accuracy is misleading and overstates performance on the majority class.

Focus on F1-score and Recall to correctly identify the minority class (readmitted).

3. DATASET OVERVIEW

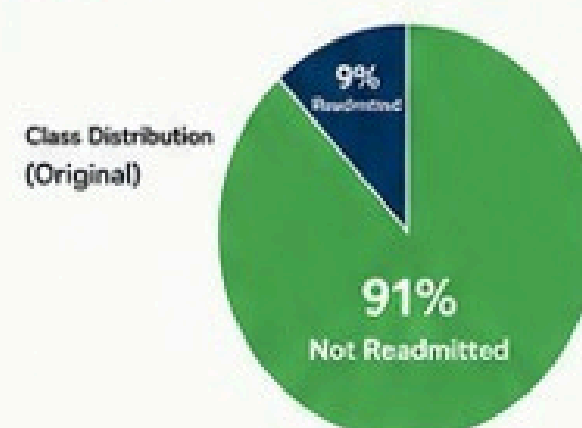
Source
UCI Diabetes 130-US Hospitals (1999–2008)

Initial Records
101,766 encounters

After Cleaning & Deduplication
71,515 unique patient encounters

Features
49 clinical and administrative features

Target
30-day readmission (Yes / No)

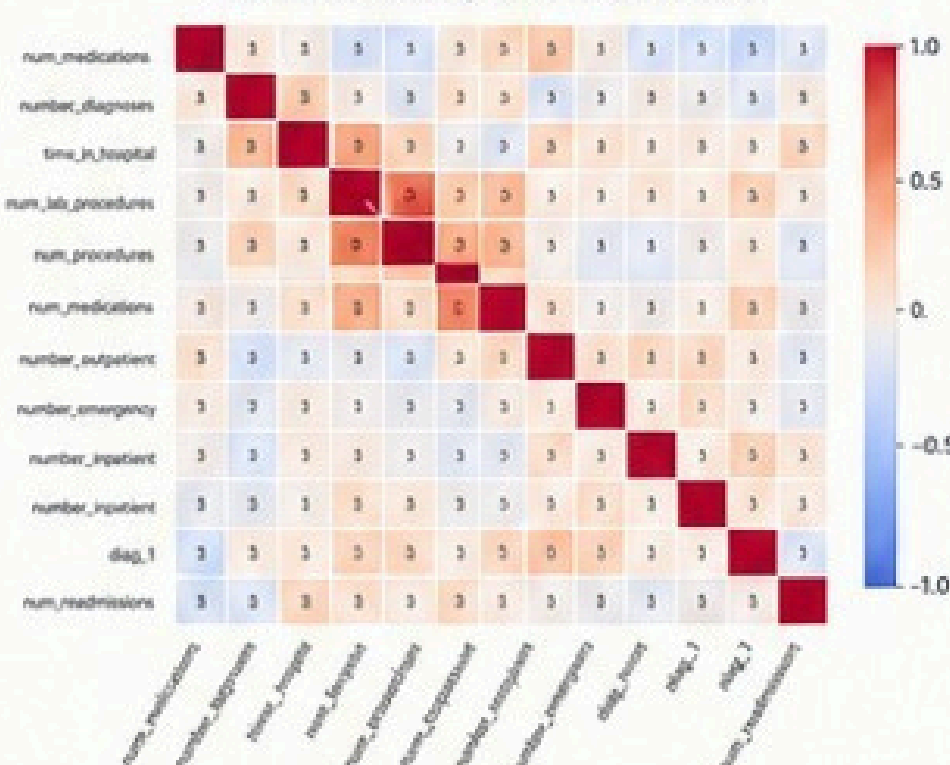


4. METHODOLOGY PIPELINE

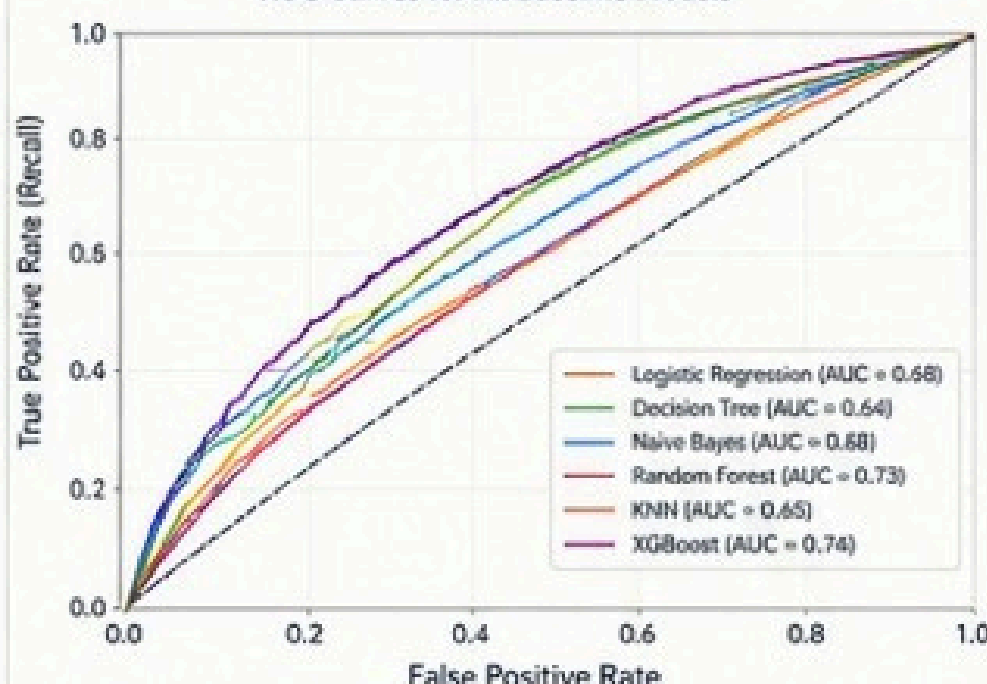
- 01 Data Loading & Inspection**
Load data, handle duplicates, explore structure & summary statistics.
- 02 Preprocessing & EDA**
Handle missing values, encode categoricals, scale features, correlation analysis & visualizations.
- 03 SMOTE Oversampling**
Address class imbalance using SMOTE to synthetically balance the minority class.
- 04 Baseline Model Training**
Train 6 different ML models and compare using cross-validation.
- 05 Model Improvement**
Hyperparameter tuning + threshold optimization to maximize F1-score.
- 06 Test Evaluation & Interpretability**
Evaluate on held-out test set, interpret model & extract insights.

5. EXPLORATORY DATA ANALYSIS

Correlation Heatmap of Numerical Features



ROC Curves for All Baseline Models



6. BASELINE MODEL RESULTS (Validation Set)

Model	Accuracy	Recall (Minority)	F1-Score (Minority)	AUC
Logistic Regression	0.74	0.16	0.32	0.72
Decision Tree	0.82	0.35	0.29	0.64
Naive Bayes	0.48	0.87	0.25	0.68
K-Nearest Neighbors	0.87	0.22	0.26	0.65
Random Forest	0.90	0.18	0.26	0.73
XGBoost (Baseline)	0.78	0.58	0.36	0.74

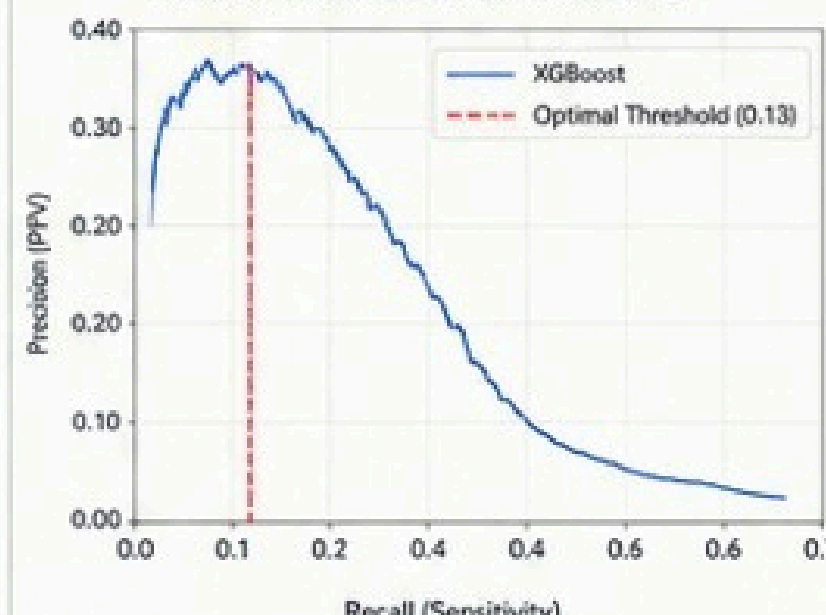
F1-SCORE COMPARISON ACROSS MODELS

XGBoost (Threshold Tuned)	0.38
XGBoost (Full Features)	0.32
Naive Bayes	0.25
Random Forest	0.26
Decision Tree	0.29
KNN	0.26
Logistic Regression	0.32
Voting Classifier	0.11
Stacking Classifier	0.07

XGBoost with threshold tuning achieved the highest F1-score (0.38) and AUC (0.72) on the held-out test set.

7. THRESHOLD TUNING (XGBoost)

Precision-Recall Curve (Validation Set)



Optimal Threshold (θ) = 0.13

Tuned threshold improves recall (sensitivity) with an acceptable trade-off in precision.

8. KEY INSIGHTS

- More medications → Higher readmission risk
- Longer hospital stay → Higher readmission risk
- Multiple diagnoses & procedures → Higher risk
- Complex patients are more likely to be readmitted

9. FINAL TEST EVALUATION (Best Model: XGBoost Tuned)

Balanced Accuracy	Recall (↑ Priority)	F1-Score (Minority)	MCC	AUC-ROC
0.68	0.58	0.38	0.22	0.72

The tuned XGBoost model substantially outperforms baseline models in identifying true readmission cases while maintaining a reasonable level of precision and overall discrimination.

10. LIMITATIONS

- Severe class imbalance remains challenging
- Precision is still relatively low at high recall
- Data noise, missing values, and coding inconsistencies
- Generalization may be limited to this dataset

11. FUTURE WORK

- Probability calibration & cost-sensitive learning
- Fairness auditing across demographic subgroups
- Incorporate temporal information & EHR sequences
- Explore transformer-based models on larger datasets
- Deploy and validate in real-world clinical settings

SCAN TO VIEW
PROJECT REPORT



INSTRUCTOR
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“Data-driven insights today,
healthier patients tomorrow.”

